

Security Assessment Report

SY-GPON-1110-WDONT

Firmware version :SYRO_3.7L_3.1.02-240517

Reserved CVE : CVE-2025-63729

Document History

Version	Author	Remark
1.0	Yashodhan Mandke	Security Report

Security Assessment Details

1.1 Executive Summary

Security Assessment of **SY-GPON-1110-WDONT version SYRO_3.7L_3.1.02-240517** model has been performed, considering below common security issues:

If firmware has any vulnerabilities

Overall security postures of the device are good, though some of the security controls/measures have not been properly thought of/implemented during the design and coding of the application. The security assessment revealed 4 High severity issue in this product.

The consolidated summary of the assessment has been presented in the Executive Summary section. Additional information is contained within the Detailed Vulnerability Information section of this report.

1.2 Scope and Objectives

The scope of this assessment was limited to Firmware of **SY-GPON-1110-WDONT version SYRO_3.7L_3.1.02-240517**. Tools used Binwalk and GNU Strings

1.3 Technology Impact Summary

The security assessments on the Firmware has been performed. These assessments aim is to uncover any security issues in the assessed router to explain the impact and risks associated with the found issues, and provide guidance in the prioritization and remediation steps. It was identified that firmware has private keys and certificates extracted from it.

Technical Findings for CVE-2025-63729

2.1 SSL Private Key extracted

Potential Impact : **High**

Description :

During the assessment it was identified that the RSA private key was hardcoded available in firmware

Affected Hosts : Device firmware, device identity, authentication.

Steps to Reproduce:

- 1 Download firmware from Syrotech website and select the product to be pen tested.



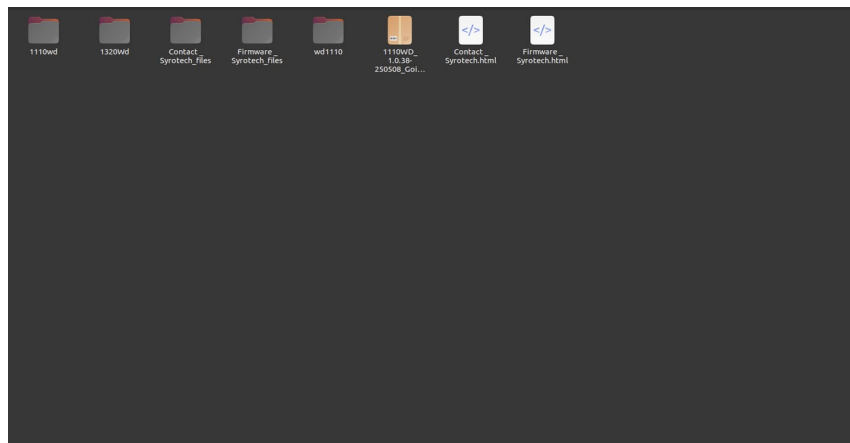
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Firmware

Stay up-to-date with our products by easily downloading the latest firmware updates. Enhance functionality and security with just a click.

Type	Model	Hardware Version	IP	Username	Password	Firmware Version	Link	Link
ONT	SY-GPON-1110-WDONT	1.0/3.2/3.21	192.168.1.1	admin	Syro@43210\$	1.0.38-250508	Download	
XPON ONT	SY-GPON-1110-WDONT	3.6/3.7	192.168.1.1	admin	Syro@43210\$	2.0.08-250224	Download	
XPON ONT	SY-GPON-1110-WDONT	3.7L/5.0	192.168.1.1	admin	Syro@43210\$	3.1.02-250124	Download	
XPON ONT	SY-GPON-2010-WADONT	4.1/4.2/4.3C/4.2E	192.168.1.1	admin	Syro@43210\$	V2.1.08-241213	Download	
XPON ONT	SY-GPON-2010-WADONT	4.3D/4.2G/4.2C	192.168.1.1	admin	Syro@43210\$	V3.2.02-240824	Download	
XPON ONT	SY-GPON-1110-WDAONT	1.3	192.168.1.1	admin	Syro@43210\$	V2.0.22-250630	Download	
XPON ONT	SY-GPON-1000-RDONT	2.8S	192.168.1.1	admin	stdONU10i / Syro@43210\$	230614	Download 1	Download 2
XPON ONT	SY-GPON-1000-2WDONT	2.2	192.168.1.1	admin	Syro@43210\$	V3.3.7-210918	Download	
XPON ONT	SY-GPON-1000-2WDONT	2.2F	192.168.1.1	admin	Syro@43210\$	V3.4.00-250630	Download	
XPON ONT	SY-GPON-4000-RDONT		192.168.1.1	admin	Syro@43210\$	1.0.10-210203	Download	
XPON ONT	SY-GPON-2000-WADONT	4.1/4.2/4.3C/4.2E	192.168.1.1	admin	Syro@43210\$	V2.1.06-240824	Download	
XPON ONT	SY-GPON-2000-WADONT	4.3D/4.2G	192.168.1.1	admin	Syro@43210\$	V3.2.00-240824	Download	
XPON ONT	SY-GPON-1100-WDONT	1.5	192.168.1.1	admin	Syro@43210\$	V1.9.1.0-230117	Download	
XPON ONT	SY-GPON-1101-WDONT	1.5	192.168.1.1	admin	Syro@43210\$	V1.9.1.1-230112	Download	
XPON ONT	SY-GPON-1100-WDONT	2.4F	192.168.1.1	admin	Syro@43210\$	V3.4.00-250630	Download	

2 Downloaded firmware



3 Extract the firmware binary using binwalk

```
essments/skytech/1110wd$ ls
1110WD_1.0.38-250508_GoipBsnlver.tar  md5.txt  _rootfs.extracted
fwu.sh                               rootfs    uImage
fwu_ver                             _rootfs-0.extracted  _uImage.extracted
stellaris@stellaris-VivoBook-ASUSLaptop-X513EAN-K513EA:~/Documents/Security_ass
essments/skytech/1110wd$
```

4 Navigate to the directory and start analyzing the extracted file system

```
essments/skytech/1110wd/_rootfs.extracted/squashfs-root$ ls
bin  dev  etc  home  image  lib  mnt  proc  sbin  sys  tmp  usr  var
```

```

essments/skytech/1110wd/_rootfs.extracted/squashfs-root/etc$ ls
app.gwdt                               inetd.conf                             runomci.sh
autoAgeingConfig.xml                  init.d                                 runsdk.sh
buildtime                             inittab                               scripts
cacert.pem                            insdrv.sh                             services
client.pem                            irf                                   setprmt_reject
config                                mdev.conf                             shells
config_default_hs.xml                 orf                                   simplecfgservice.xml
config_default.xml                     passwd                                solar.conf
custname                              ppp                                   solar.conf.in
dhclient-script                       product                               ssl_cert.pem
dnsmasq.conf                          profile                               ssl_key.pem
dropbear                              radvd.conf                           TZ
ethertypes                            rc_voip                              version
ETSI_T3R10.BIN                        resolv.conf                           voip_init.xml
fw_dxs-2.1.3.bin                       rtk_tr142.sh                         voip_lan.xml
gn25l95_eeprom.bin                    rtl-3320-backup.bin                  wscd.conf
hosts                                  runoam.sh

```

- 5 Extract from binwalk on the internal file

```

essments/skytech/1110wd/_rootfs.extracted$ ls
0.squashfs  squashfs-root  squashfs-root-0

```

- 6 In the binwalk output one of the finding is SSL private key

```
-----BEGIN PRIVATE KEY-----
MIIEvQIBADANBgkqhkiG9w0BAQEFAASCBAcwggSjAgEAAoIBAQDPIDfMcXHv/iAh
Ic7+0YuXtj+rE7nXUeQ1zbQNRR49++krY2KsHjrrbxn11MYtJWywhuYPPVBdaTJn
cHAfkZ6pCihRceTMID7q1Rd96v6GNbxwPeCBEP1BeE0HOAENUucmkH7ct9a36age
6qyZ5cb7YINbm6F35ZCMyI6bBddQ+MY0dHt/o57LWrl07kpNj1P1yXpZpYkwcugp
xrMhgqGp1NGLpu+VzhG/dwHyEk2+s/TAmikn2yewqkLJp4KvPaUUF013T/jY2Ay8
c5H3JmE5mpCAG0s1P+IQyj/vP+6Wqx9e51DkwriinX+xgs21mgfiC0omDbopgGNb
XCDXG5IPAgMBAAECggEAff6BMtIHLw+lB8KDIirhmWX5kN2s/LgJ9FivICtdXCMz
EXuheEXjXXf1+L531tqfJTrKE5cendPyH04cRnfUZcuR/wkjPrCW1U70vYAHWCiz
+isJuDlXdU3yvWnp8GrvxVu8cBM8gB+Yb0YdvHgZ9JIffci0DjUNGzac2NxSwC0c
CJssh87dr+2t4/8bDWIYlmLRazyLpPJ7gxwzjwQnHGEi+0WUBzvRJZmJHkiCdAhM
Xr7+yunr9y0EJ66DrdX0bQvur8vzCQdsG2CpvYPMJXB0TUIzkfNTYrD8DjrXtV51
q69Wy+gYZB8eL9E58cKnjt6jJmSGN0e070xDAT3E8QKBgQDx+WTqzLmvHR+GWIMr
J/16Aid6pNFMdIDGzJm5vUofK/s99Yd7RaI5U/1CAfFE7uYoNG04Vm0eNY6WZUw
Yv1cR5kmSGzCfBkRqf2VEIQCha/dpbNrArKMg9UcXHGAokjaJPRfbjZ2B4IOfUbg
LtyH2kztZd0fF5iwyHc9cKYKGwKBgQDbIbddaQ61hbPmIz8+P9oeh8Sf+V9cE1lS
Mh1ndhMLuo6q1QL5tVzJp3K4UsDFSjS7ehYecjFrZxgKkyrRNdsusj4vQ8dwvYp7
```

2.2 RSA Key extracted

Potential Impact : **High**

Description :

During the assessment it was identified that the RSA private key was hard coded available in firmware

Affected Hosts : Device firmware, device identity, authentication.

Steps to Reproduce:

- 1 Continue analyzing same file extracted by gnu strings
- 2 EC private key is available in same file. There is keys of RSA


```
-----BEGIN RSA PRIVATE KEY-----
Proc-Type: 4,ENCRYPTED
DEK-Info: DES-EDE3-CBC,570ECFA93805BE9F

dDc704a7oeh0AaRLEw9M593i3YYb+QoEyxt+Q+tdw7Puy3dCfa/CNSUSF81SqcGa
bbunmRbDP2/p+sPl18u+oLQIDDvQ+TDF0IgGLq9T7k1+YqUm92gmT08XZ8j/PCiF
ANJYa25qJhjcvI9U1U66cjBc+kbYazJZF1MqnnIVlt5kHp4axbR4Bxpm2Wco0El9
cZFKPq6gZnBTs7XzLNp6JShUtF1bEv6ZnxJy1sW1neLwYCUaWBA4kBUH6TL67gLu
CM49USepIiXINsH1X0yt6/PskLA/Gyj+7gBmlnkoQqr1E9sUvDFqp1FepyW43231
Rex3tDJKKrF4XCqMrDANGTNvdFGSknC+TRkMRD17Z2Gy2sPVaAu0m4oU5F/pgv0W
h3r3Z1yjJPchtxhhqFBGYUqshLaSZxaG/UhKP7vPKVUk7uRuLfK/zpQKn0jkm/3t
gYrwMioow5Avs2qbg2qjNkNLk4mGvWwrTEJoD0gjdu52xEhyDZrtIMkD8B3Z/Awe
0o0r1p4AKi8wgSWsv0rja3fnuNXT7MCf6XzV3gRFRUtcc/FqmkgSFSr0Y0gomKL0
ETSPJdGEwx2+gr+k+UywDSHr6Q9Q9nTMzC/0tDKPdFchnuwsJpJbILAsByuSxCyH
MxsQStJsqb7PahtjAaFBXEfWJpMCIFwPyj/vygdLm/EMMmTPMUTR2KlNioCsUtlp
zilu2Hrg5iJRzdDacdEeMZqqnEhPduis3Xm74bg2PP08IzBgSaWU00k6lxKuC09A
2nih/rRWSDGU1wQoao9CoUl9ZPykd56l84oCHfVvA9dKSmj1Eht7Wg==
-----END RSA PRIVATE KEY-----
```

2.3 Client Certificate and CA Certificate

Potential Impact : **High**

Description :

During the assessment it was identified that the second RSA private key was hard coded available in firmware

Affected Hosts : Device firmware, device identity, authentication.

Steps to Reproduce:

- 1 Continue analyzing same file extracted by binwalk.
- 2 Client certificate is available in same file.

-----BEGIN CERTIFICATE-----

MIIDgDCCAumgAwIBAgIBADANBgkqhkiG9w0BAQQFADCBjTELMAkGA1UEBhMCVFcx
DzANBgNVBAGTBIRhaXdhbjEQMA4GA1UEBxMHSNpbkNodTEQMA4GA1UEChMHUmVh
bHRlazePMA0GA1UECXMgQ04gU0Q1MRUwEwYDVQQDEwNzIuMjEuNzAuNDExITAfBgkqhkiG9w0BCQEWEmd1ZXN0QDE3Mi4yMS43MC40MTAeFw0wNjEyMDUwODMzMzNaFw0w0TEyMDQwODMzMzNaMIGNMQswCQYDVQQGEwJUVzEPMA0GA1UECBMGVGFpd2FuMRAwDgYDVQQHEwdIc2luQ2h1MRAwDgYDVQQKEwdSZWFsdGVrMQ8wDQYDVQQLEwZDTiBTRDUxFTATBgNVBAMTDDE3Mi4yMS43MC40MTEhMB8GCSqGSIb3DQEJARYSZ3VLc3RAMTcyLjIxLjcwLjQxMIGfMA0GCSqGSIb3DQEBBAAQ4GNADCBiQKBgQCurM237vPQNipvM4c3E9SgLUlreKisXsQpmQAQot2NJRr+h6zc3Y2jkx9MveweYrFpMtDYpq7ojkC+5q1baa2ySUplxlEnGTGfU0lbgKfQyd5SixI+UQIRXgV/7005NPgwPGehfbXnK5LcWHfr8ZcJhDNCUL77V/9cqDyQhSInGwIDAQABo4HtMIHqMB0GA1UdDgQWBBS850K9CgQ77nbyrY2exn7+DGngrzCBugYDVR0jBIGyMIGvgBS850K9CgQ77nbyrY2exn7+DGngr6GBk6SBkDCBjTELMAkGA1UEBhMCVFcxDzANBgNVBAGTBIRhaXdhbjEQMA4GA1UEBxMHSNpbkNodTEQMA4GA1UEChMHUmVhbHRlazePMA0GA1UECXMgQ04gU0Q1MRUwEwYDVQQDEwNzIuMjEuNzAuNDExITAfBgkqhkiG9w0BCQEWEmd1ZXN0QDE3Mi4yMS43MC40MYIBADAMBgNVHRMEBTADAQH/MA0GCSqGSIb3DQEBBAAUA4GBAKbF7ugvAfxe0FJTWJppnRv7rZJmzblzwaUhXStvqcgpdj3JLwwjdX+QSYSVk5GGWzNCYcAeXSMnF9k3M4lu7V/LIa+AeoL9mg4KlMj500AtDRttv7J0oPnEIID30+YvuseLQGgX9MhfRUt23hI70hHhRuknjFmpeb04PeywYHRE

-----END CERTIFICATE-----

```

-----BEGIN CERTIFICATE-----
MIIDaTCCAIECFA75yHUy22x2RfpkqHuj4BXtjNcQMA0GCSqGSIb3DQEBCwUAMHEx
CzAJBgNVBAYTAkFVMRMwEQYDVQQIDApTb211LVN0YXRlMRwwGgYDVQQKDBNEZWZh
dWx0IENvbXBhbnkgTHRkMRwwGgYDVQQLDBNEZWZhdWx0IENvbXBhbnkgTHRkMREw
DwYDVQQDDAh0ZXN0Lm9yZzAeFw0yNDA0MjgwMTQyMzVaFw0yOTA0MjcwMTQyMzVa
MHExCzAJBgNVBAYTAkFVMRMwEQYDVQQIDApTb211LVN0YXRlMRwwGgYDVQQKDBNE
ZWZhdWx0IENvbXBhbnkgTHRkMRwwGgYDVQQLDBNEZWZhdWx0IENvbXBhbnkgTHRk
MREwDwYDVQQDDAh0ZXN0Lm9yZzCCASIwDQYJKoZIhvcNAQEBBQADggEPADCCAQoC
ggEBAM8gN8xxce/+ICEhzv7Ri5e2P6sTuddR5DXNtA1FHj376StjYqwe0utvGfXU
ki0lbLCG5g89UF1pMmdwcB+RnqkKKFFx5MwgPurVF33q/oY1vHA94IEQ/UF4TQc4
AQ1S5yaQfty31rfpqB7qrJnlxvtgg1uboXflkIzIjpsF11D4xg50e3+jnstauU7u
Sk2PU/XJelmliRZy6CnGsyGCoanU0Yum75X0Eb93AfISTb6z9MCAkSfbJ7CqQsmn
gg89pRQU7XdP+NjYDLxzkfcmYTMakIAbSzU/4hDKP+/8/7parH17nU0TCuKKdf7GC
zbWaB+ILSiYNUimAY1tcINcbkg8CAwEAATANBgkqhkiG9w0BAQsFAA0CAQEAEPVb
JtH5Cwtxlzapxi14PHV2ceil/TMzbIamn8L9lt7IoDHKhGsZ1QviXNvlhjPF6WtL
DNEpq6Uwj9fG2ycPtsBIxkmEXKEU+fC9r1PAbt7/xYkjYkYWh02uu1kwiBczKsbs
Bnbm7h+96ay5ptnuhWxTUu9rGIC2eQfN6Y1fWBjABWQ+e8IvBZZuxT3V6CvjFYai

```

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